Web Design - Computer lab 6

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Intro to JavaScript: The "Old Way"

Avoid Inline JavaScript: Historically, logic was written directly inside HTML attributes. This breaks the **Separation of Concerns** and makes code hard to maintain.

```
<p id="demo">JavaScript can change HTML content.</p>
<button
    type="button"
    onclick='document.getElementById("demo").innerHTML = "Hello JavaScript!"'>
    Click Me!
</button>
```

Listing: Anti-pattern: Mixing HTML and JS

```
<img id="myImage" src="pic_bulboff.gif" style="width:100px">
<button
    onclick="document.getElementById('myImage').src='pic_bulbon.gif'">
    Turn on the light
</button>
```

Listing: Anti-pattern: Inline Event Handlers

Separation of Concerns: The Modern Way

Best Practice: Keep HTML (structure) and JS (behaviour) in separate files. Load your script using the **`defer`** attribute in the `<head>`.

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Modern JS</title>
    <!-- defer ensures the HTML parses before the JS runs -->
    <script defer src="app.js"></script>
</head>
<body>
    <h2 id="title">My First Web Page</h2>
    <p id="demo">My first paragraph.</p>
    <button id="actionBtn">Click Me</button>
</body>
</html>
```

Listing: index.html

I use jQuery
What does jQuery mean?
It means he's afraid of the DOM
No, it doesn't.
document.getElementById
Stop it, Patrick, you're scaring him
ProgrammerHumor.io

Task 1: Sticky Menu

1. Sticky menu
   - The main menu remains at the top of the screen while scrolling.
   - **Challenge:** Implement this behaviour using JavaScript instead of the CSS `position: sticky` property.
   - **Hint:** Create a `.sticky` CSS class and toggle it using a scroll event listener.

```
const navbar = document.getElementById("menu");
const stickyOffset = navbar.offsetTop;

window.addEventListener("scroll", () => {
    // Check how far the user has scrolled
    if (window.scrollY >= stickyOffset) {
        navbar.classList.add("sticky");
    } else {
        navbar.classList.remove("sticky");
    }
});
```

Task 2: Smooth scroll (Setup)

```
html {
    scroll-behavior: smooth;
}
```

Listing: The Modern CSS Way

1. Call function on any a-href click

```
document.querySelectorAll('a(hrefˆ="#")').forEach(anchor => {
    anchor.addEventListener('click', function(event) {
        // Prevent instant jump
        event.preventDefault();

        // Pass the event to our custom scroll function
        scrollFunction(event);
    });
});
```

Listing: JavaScript Event Listener

Task 2: Smooth scroll (Function)

2. Scroll function

```
function scrollFunction(event) {
    const targetId = event.currentTarget.getAttribute('href');
    const target = document.querySelector(targetId);

    // Smooth scroll to the target section
    if (target) {
        target.scrollIntoView({
            behavior: 'smooth'
        });
    }
}
```

Listing: JavaScript Scroll Logic

Task 3: Hamburger Menu

1. For screens smaller than 576px, the main menu will be hidden and replaced by a single hamburger icon.
2. Clicking this icon will open/close the entire menu.
3. **Hint (CSS):** Use `@media (min-width: 576px)` to hide the hamburger icon on desktop screens. By default on mobile, the menu class should have `display: none`.
4. **Hint (JS):** Use an Event Listener to toggle an **`.opened`** class, just like you did in Task 1.

```
const menuBtn = document.getElementById("btnMenu");
const navBlock = document.getElementById("blockNavigation");

// Do not use inline onclick! Use an event listener:
menuBtn.addEventListener("click", () => {
    // toggle() adds the class if missing, removes it if present
    navBlock.classList.toggle("opened");
});
```

Listing: Vanilla JS Toggle

Task 3: Hamburger Menu

What will be evaluated?

1. The site has a sticky menu
2. Scrolling through the sections is smooth
3. On mobile resolution, there is a functioning hamburger menu
4. **Validate** your solution (zero warnings/errors) using:
   - HTML validator
   - CSS validator

JS Frameworks

- Angular demo (https://angular.dev/playground)
- VueJS (https://play.vuejs.org)
- React (https://reactplayground.vercel.app/)